



YouTube Learning State Layer

From watching what you want — to learning what you need

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 Assignment 2 - Reimagining a Product With AI & Prototyping

Problem & Context



Current Scenario -

India's exam culture isn't just a trend — it's a tradition. With the majority of its 1.4 billion population concentrated in Tier 2 and 3 cities, **academics remain the most accessible and certain path to upward mobility**. This belief, passed down through generations, has created the world's most competitive exam ecosystem. Numbers prove this -

- 24 lakh+ students registered for NEET 2024 alone.
- 14 lakh+ students appeared for JEE Main 2024.
- 13 lakh+ applied for UPSC 2024 — with only 1,056 final selections.



What is the Problem ?

YouTube was built for entertainment — not education. Yet for millions of Indian exam aspirants who cannot afford private coaching, it has become their classroom, their teacher, and their only resource.

The past decade saw an explosion of free, high-quality teaching content on YouTube — numerous educators democratized exam preparation for Tier 2 and 3 India. But the platform hosting all of this never evolved with its users.

YouTube **has no memory of your syllabus, no awareness of your progress, and no ability to bring you back in context after a gap.**



Users Impacts on solving -

- Reduces exam anxiety.
- Restores confidence and direction.
- **Eliminates wasted revision time.**
- **Democratizes structured learning.**
- Reduces video/playlist dropout.
- Improves Learning Retention.

JTBD & User Persona

Job To Be Done -

→ When I sit down to study on YouTube, I want to feel oriented and in control of my learning path, so I can maximize every study hour in a high-stakes, time-sensitive preparation.

“Jab bhi YouTube pe padhai shuru karta hoon uske do-teen ghante baad pata chalta hai ki sirf do videos dekhi, vo bhi aadhi-aadhi 😞😞 - **there is absolutely no structured and meaningful progress !**”

Goals/Motivations -

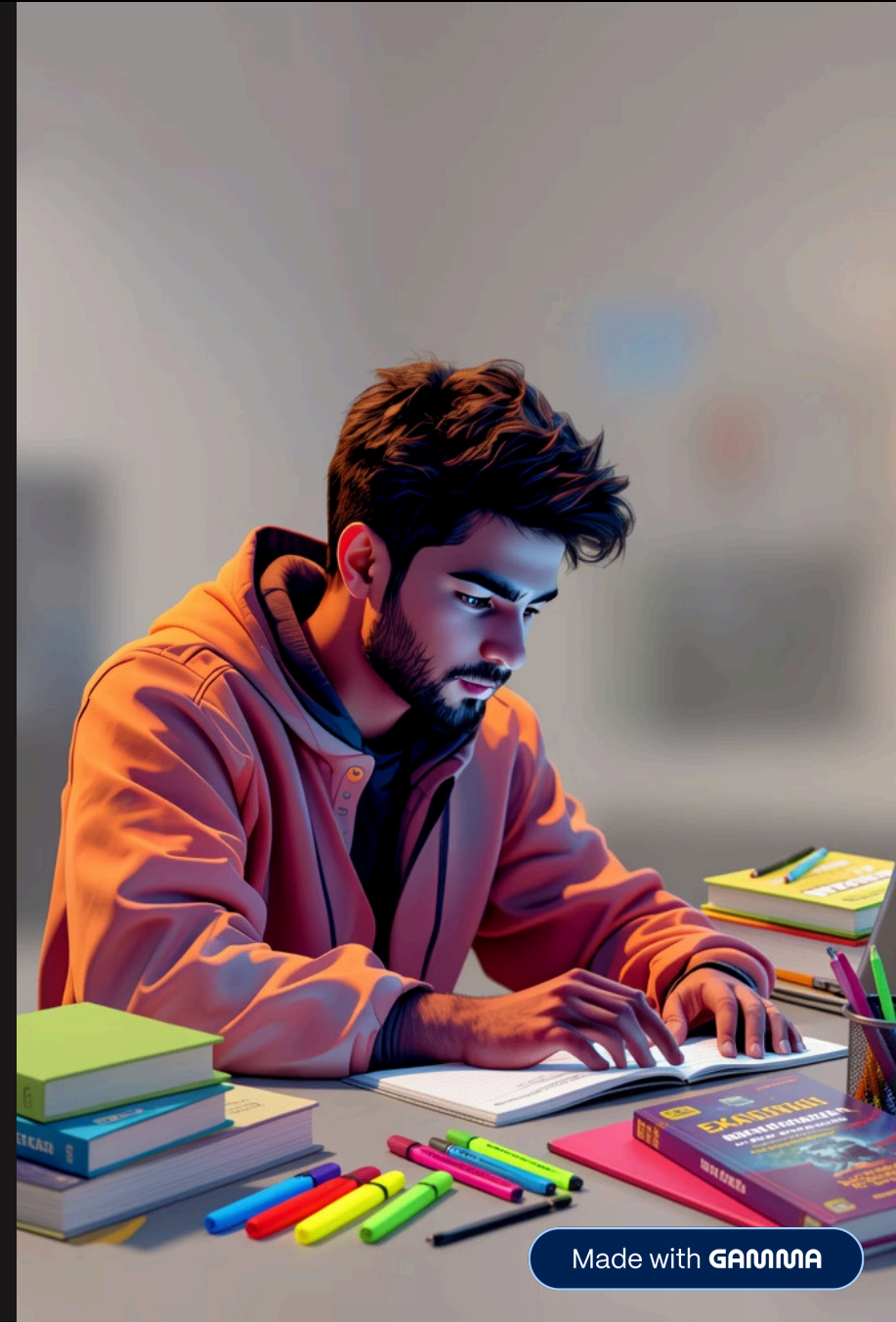
Aarav Sharma - 16 year old boy from Lucknow. His goal is to crack NEET and secure a medical seat — not just for himself, but to validate every sacrifice his family has made. Every study hour feels like it either brings him closer or pushes him further from that one goal

Workflow -

Weekdays structured around Physics Wallah lectures — 4 to 5 hours across Physics, Chemistry, Biology — followed by long problem-solving sessions from reference books. YouTube enters the routine occasionally but purposefully, for concept revision or doubt resolution. Relies on physical notes but rarely makes them consistently. Watches educational videos passively, like movies.

Pain Points -

- Returns to a playlist after days and has no idea what he already covered
- Wastes precious time rewatching content he's seen before
- Feels like he's watching but not actually progressing through the syllabus
- Every wasted hour triggers anxiety about the exam timeline



How People Solve This Today ?

The Workarounds Are Broken Too ✖

What Students Do Today:

When continuity breaks, students have two choices — neither works well.

Rewatch from Video 1

Emotionally connected to their channel, or simply unsure where they left off. Time wasted, motivation drained.

Abandon and start fresh

Open a new playlist, new channel, new topic. The original learning journey dies quietly.

Both behaviours have the same outcome: **the syllabus gap widens.**

What Existing Tools Offer:

Tools do exist — but none solve the real problem:

TrackMyCourse

Adds checkmarks and progress bars to playlists. Tracks what you watched. No syllabus awareness, no AI, no re-entry context.

YouTube Learning Tracker

Tracks watched videos and learning plans. Still watch-state only. No concept-level intelligence.

StudyTrackYT

Adds AI quizzes and notes. Closest to our space but no syllabus mapping, no re-entry summaries, not built for Indian exam context.

The Gap None of Them Fill:

Every existing solution tracks **watch progress**.
None track **learning progress**.

- ❑ None know you're preparing for NEET. None know you're on Unit 3 of Physics. None generate a re-entry summary after a 10-day gap. None recommend the next video based on your syllabus position.

That gap is exactly where we live.

Problem Validation

Research Method:

Conducted informal interviews with 4 Indian competitive exam students aged 16-20 across NEET and JEE preparation cohorts. Questions focused on YouTube learning behaviour, drop-off patterns, return behaviour, and honest note-taking habits. Examination of playlist view patterns, studying comments, reviews on app store and play store.

Key Findings -

- **1 in 3** playlists started are never finished
- **45-50 mins** is the real drop-off threshold — not the 1.5 hours users claim
- **0 out of 4** students actually made consistent physical notes despite claiming otherwise
- **"I changed my interest"** — identified as a face-saving response masking loss of context
- **Millions of views on video 1. Thousands on video 6.** YouTube's own data confirms the drop-off is real and universal

"Users don't abandon playlists because they lose interest. They abandon them because the platform gives them no reason to return."

The problem isn't student discipline. The problem is a product that has no memory.

Why this matters to the business ?

- **491 million YouTube users in India** — its single largest market globally, nearly double the US
- **₹16,000 crore** contributed by YouTube's creative ecosystem to India's GDP in 2024 alone
- **\$36.1 billion** in global ad revenue in 2024 — a 14.6% YoY increase — built entirely on watch time and engagement
- **50M+ exam aspirants** using YouTube as their primary free learning resource — highest-intent users on the platform
- Every abandoned playlist = lost watch time = lost ad revenue. Improving completion among this segment directly compounds YouTube's most critical business metric
- **The opportunity:** Convert India's most motivated learners from one-time viewers into returning, progressing users — watch time, session depth, and revenue follow automatically

Brainstorming Solutions

Why a Browser Extension ?

The Four Approaches Considered:

Mobile App	High capital, large engineering team, significant infrastructure investment — not viable at this stage
YouTube Feature	Google is already revamping YouTube's AI integration internally — introducing a new feature into that chaos risks getting lost, delayed, or overridden entirely
Web App Companion	Creates cognitive overload — forces students to switch tabs mid-study, adding friction to an already fragile learning moment
Browser Extension	✓ Technically feasible, zero platform dependency, lives directly on YouTube, zero context switching

Why Browser Extension Wins ?

- **Zero friction** — sits invisibly on YouTube, activates only when needed, never asks the student to go anywhere else
- **No platform permission needed** — doesn't require YouTube's cooperation or API access. We build on top, not inside
- **Proven form factor** — tools like Glasp and VideoSummarizer already validate that students accept browser extensions as YouTube companions

✓ *The best solution is the one that meets the student exactly where they already are — on YouTube, mid-session, with no time to spare.*

A browser extension is the only approach that does this without asking anything extra from the user.

Solution Overview

Introducing — YouTube Learning State Layer

What It Is: A browser extension that remembers what you learned — not just what you watched. YouTube has watch state. We built learning state — a persistent, syllabus-aware memory layer that sits on top of YouTube and tracks concept-level progress for every Indian exam aspirant.

Four AI Capabilities Powering It:

1

Categorization & Classification

NLP models read video titles, descriptions, and transcripts — mapping each video to its exact syllabus topic.

2

Concept-Level Progress Tracking

ML inference from behavioural signals — rewinds, drop-off points, playback speed, session duration.

3

Contextual Re-entry Summary Generation (GenAI)

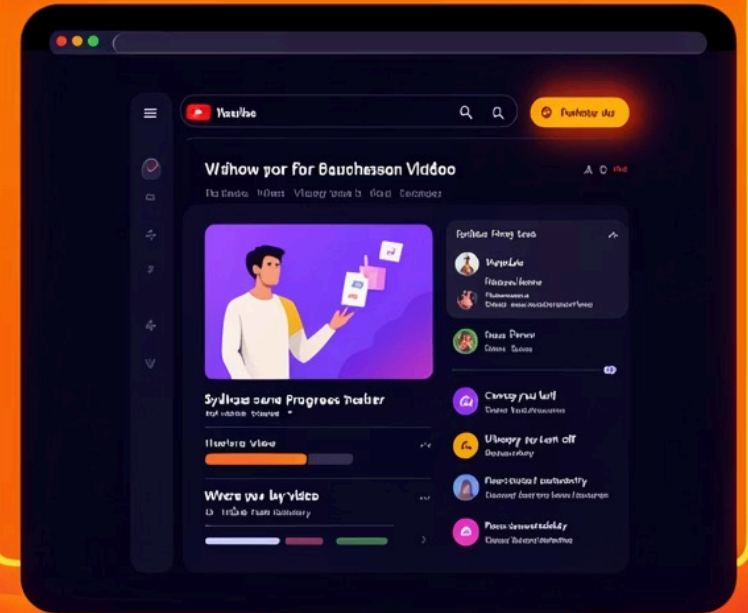
When a student returns after a gap, a fresh, personalized natural language summary is generated — what was covered, where they left off, what's coming next.

4

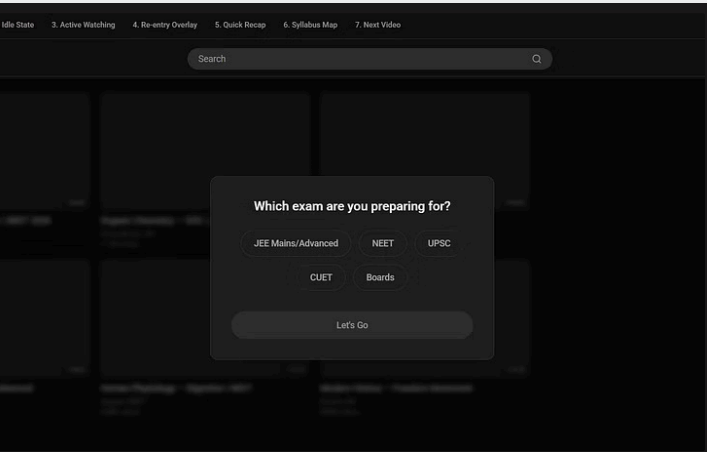
Progression-Based Recommendation (Ranking)

Next video recommendations driven purely by syllabus sequencing and identified learning gaps — not YouTube's engagement algorithm.

Link to Working Prototype - [click here](#)

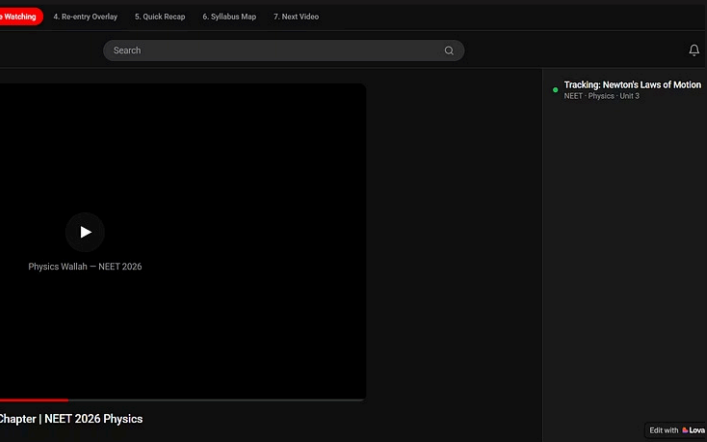


User Work-Flow



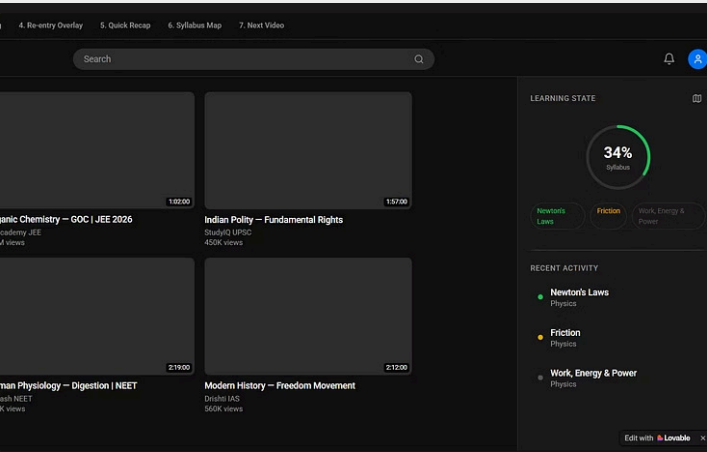
S1 — Onboarding

Aarav selects NEET on first install. Done in 10 seconds. Never appears again.



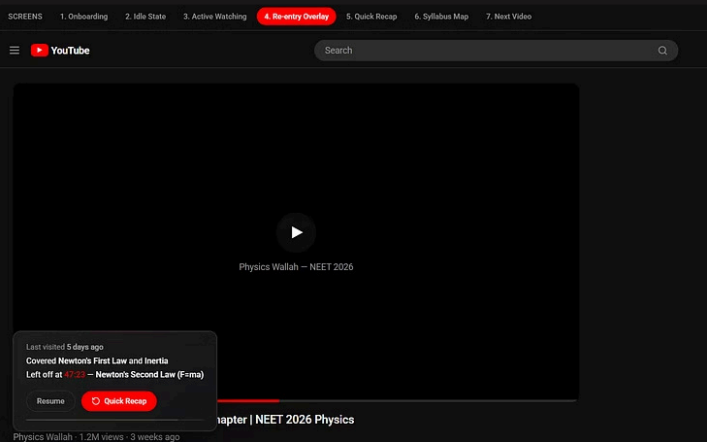
S3 — Active Watching

Single tracking line — topic name and unit. Silent. Minimal. Extension is working invisibly.



S2 — Idle State

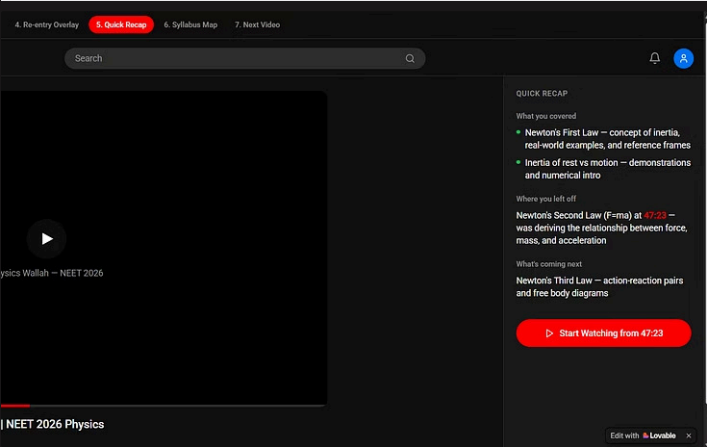
Sidebar shows syllabus progress ring and recent activity. He knows where he stands before pressing play.



S4 — Re-entry Overlay

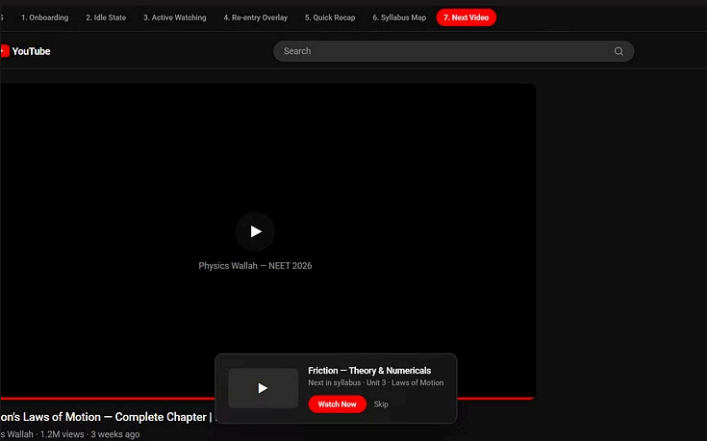
Returns after 10 days. Floating card surfaces — last visited, topics covered, drop-off point. Resume or Quick Recap.

User Work-Flow



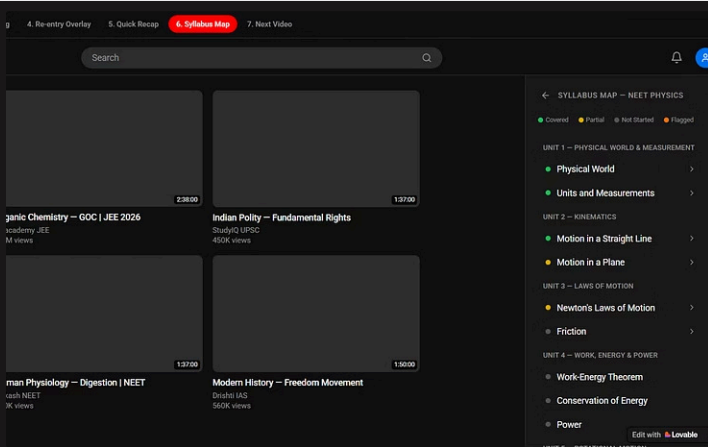
S5 — Quick Recap

AI generates what was covered, where he left off, what's next. One button — Start Watching from 47:23.



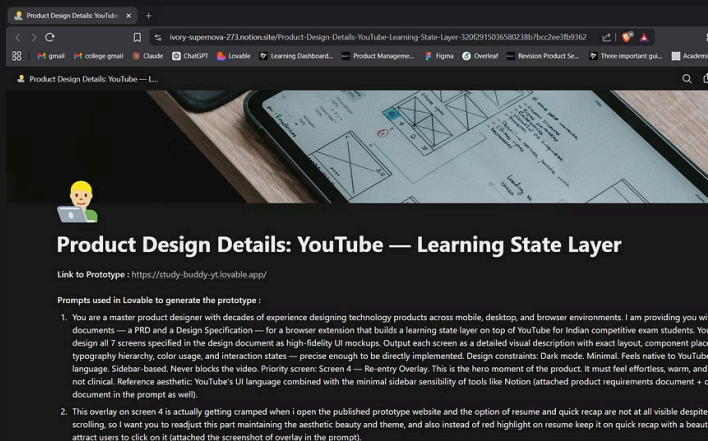
S7 — Next Video

Video ends. Extension recommends next syllabus-sequenced video. Not YouTube's algorithm. His learning path.



S6 — Syllabus Map

Full color-coded syllabus tree. Green, yellow, grey, orange. He sees his entire NEET journey at a glance.



Link to Prompting Details Document -

[click here](#)

Metrics and Pitfalls

North Star Metric:

 **Playlist Completion Rate** % of exam-prep playlists started that are completed meaningfully — defined as 80%+ of videos watched.

Secondary Metrics:

 **Return Rate** — % of dropped playlists returned to within 7 days.

 **Time to Re-entry** — average days between drop-off and return.

 **Restart Rate** — % of returning students who restart from video 1.

 **Session Depth** — average videos watched per study session.

Guardrail Metric:

 **Passive Completion Rate** % of students completing playlists without meaningful watch time per video.

Pitfalls ⚠️	Mitigations ✅
1.) Syllabus Misclassification - AI maps wrong topic → wrong progress tracked	Launch with pre-verified trusted channel playlists first
2.) Re-entry Summary Ignored - Students dismiss overlay without reading	Keep under 4 lines, auto-dismiss in 10 seconds, A/B test formats
3.) Low Extension Retention - Students install and abandon within a week	Deliver syllabus map value on first install before first drop-off occur

Link to PRD - [click here](#)